
End trigger — bar / knee height
bar estimate (wrist + ⅙ forearm)
knee
px from bottom
1400
1200
1000
end trigger: bar clears knee
Start trigger — wrist-centroid speed (9-sample mean, ±2 baseline)
torso-lengths / s
3
2
1
0
no still window before the crossing:
clip starts mid-pull, start clamped to frame 0
onset 0.15 L/s
Measurement — trunk angle to the floor, mean of L/R (band = L/R disagreement)
degrees
80
60
40
first pull
mean 47.6°
SD 11.8°
drift -30.5°
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
video time (s)